

DBMS EXPERIMENT

1) Stored procedure

```
mysql> CREATE DATABASE college_db;
Query OK, 1 row affected (0.10 sec)

mysql> USE college_db;
Database changed
mysql> CREATE TABLE student (
    -> id INT PRIMARY KEY ,
    -> name VARCHAR(50),
    -> marks INT
    -> );
Query OK, 0 rows affected (0.09 sec)

mysql> INSERT INTO student VALUES
    -> (1,'arun',85),
    -> (2,'priya',72),
    -> (3,'rahul',90),
    -> (4,'divya',65);
Query OK, 4 rows affected (0.03 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM student;
+----+-----+-----+
| id | name  | marks |
+----+-----+-----+
| 1  | arun  | 85    |
| 2  | priya | 72    |
| 3  | rahul | 90    |
| 4  | divya | 65    |
+----+-----+-----+
4 rows in set (0.01 sec)
```

```
-> (4,'divya',65);
Query OK, 4 rows affected (0.03 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM student;
```

id	name	marks
1	arun	85
2	priya	72
3	rahul	90
4	divya	65

```
4 rows in set (0.01 sec)
```

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE show_students()
-> BEGIN
->     SELECT * FROM student;
-> END //
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> DELIMITER ;
mysql> CALL show_students();
```

id	name	marks
1	arun	85
2	priya	72
3	rahul	90
4	divya	65

```
4 rows in set (0.00 sec)
```

```
Query OK, 0 rows affected (0.01 sec)
```

2) FUNCTION

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement

```
mysql> USE college_db ;
```

Database changed

```
mysql> SELECT * FROM student;
```

id	name	marks
1	arun	85
2	priya	72
3	rahul	90
4	divya	65

4 rows in set (0.00 sec)

```
mysql> DELIMITER //
```

```
mysql>
```

```
mysql> CREATE FUNCTION get_grade(marks INT)
```

```
    -> RETURNS VARCHAR(10)
```

```
    -> DETERMINISTIC
```

```
    -> BEGIN
```

```
    ->     IF marks >= 80 THEN
```

```
    ->         RETURN 'A';
```

```
    ->     ELSEIF marks >= 60 THEN
```

```
    ->         RETURN 'B';
```

1	arun	85	
2	priya	72	
3	rahul	90	
4	divya	65	

4 rows in set (0.00 sec)

```
mysql> DELIMITER //
```

```
mysql>
```

```
mysql> CREATE FUNCTION get_grade(marks INT)
```

```
    -> RETURNS VARCHAR(10)
```

```
    -> DETERMINISTIC
```

```
    -> BEGIN
```

```
    ->     IF marks >= 80 THEN
```

```
    ->         RETURN 'A';
```

```
    ->     ELSEIF marks >= 60 THEN
```

```
    ->         RETURN 'B';
```

```
    ->     ELSE
```

```
    ->         RETURN 'C';
```

```
    ->     END IF;
```

```
    -> END //
```

ERROR 1304 (42000): FUNCTION get_grade already exists

```
mysql>
```

```
mysql> DELIMITER ;
```

```
mysql> SELECT name,marks,get_grade(marks) AS grade FROM student;
```

name	marks	grade
arun	85	A
priya	72	B
rahul	90	A
divya	65	B

4 rows in set (0.00 sec)

```
mysql> |
```

3)CURSOR

```
mysql> USE college_db;
Database changed
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE cursor_example()
-> BEGIN
->     DECLARE done INT DEFAULT 0;
->     DECLARE sname VARCHAR(50);
->
->     DECLARE cur CURSOR FOR
->         SELECT name FROM student;
->
->     DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
->
->     OPEN cur;
->
->     read_loop: LOOP
->         FETCH cur INTO sname;
->
->         IF done = 1 THEN
->             LEAVE read_loop;
->         END IF;
->
->         SELECT sname AS student_name;
->     END LOOP;
->
->     CLOSE cur;
-> END //
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
```

```
Query OK, 0 rows affected (0.02 sec)
```

```
mysql>
```

```
mysql> DELIMITER ;
```

```
mysql> CALL cursor_example();
```

```
+-----+  
| student_name |
```

```
+-----+
```

```
| arun          |
```

```
+-----+
```

```
1 row in set (0.01 sec)
```

```
+-----+  
| student_name |
```

```
+-----+
```

```
| priya         |
```

```
+-----+
```

```
1 row in set (0.01 sec)
```

```
+-----+  
| student_name |
```

```
+-----+
```

```
| rahul         |
```

```
+-----+
```

```
1 row in set (0.01 sec)
```

```
+-----+  
| student_name |
```

```
+-----+
```

```
| divya         |
```

```
+-----+
```

```
1 row in set (0.01 sec)
```

```
Query OK, 0 rows affected (0.02 sec)
```

4)TRIGGER

```
Query OK, 0 rows affected (0.02 sec)

mysql> USE college_db;
Database changed
mysql> CREATE TABLE student_log (
    ->     message VARCHAR(100)
    -> );
ERROR 1050 (42S01): Table 'student_log' already exists
mysql> USE college_db;
Database changed
mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER student_insert_trigger
    -> AFTER INSERT ON student
    -> FOR EACH ROW
    -> BEGIN
    ->     INSERT INTO student_log
    ->     VALUES ('New student added');
    -> END //
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
mysql> INSERT INTO student
    -> VALUES (6,'Kumar',80);
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM student_log;
+-----+
| message                |
+-----+
| New student added     |
+-----+
1 row in set (0.00 sec)

mysql> |
```

5) CONTROL STRUCTURES

```
mysql> USE colleg_db;
ERROR 1049 (42000): Unknown database 'colleg_db'
mysql> USE college_db;
Database changed
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE check_marks(IN m INT)
    -> BEGIN
    ->     IF m >= 50 THEN
    ->         SELECT 'PASS' AS Result;
    ->     ELSE
    ->         SELECT 'FAIL' AS Result;
    ->     END IF;
    -> END //
Query OK, 0 rows affected (6.10 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL check_marks(80);
+-----+
| Result |
+-----+
| PASS   |
+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

mysql> |
```